

Apply to be a Science Corps Fellow - Entry #3376

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Female
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United States
When did you or do you expect to complete your PhD?
Spring 2026
Please tell us briefly about your university education (institution, city, country, year)
CUNY Hunter College: B.A in Chemistry (New York, NY, U.S., 2015-2019) Washington University in St. Louis: MS and PhD in Biomedical Engineering (2019-ongoing)
Why would you like to be a Science Corps Fellow?
I would like to be a Science Corps Fellow because I have personally experienced the transformative impact of STEM outreach and mentorship, and I want to extend those opportunities to students in resource-limited communities. My own trajectory in science began with participating in STEM outreach programs that opened doors I otherwise would not have known existed. As a high school student, I participated in the Urban Barcode Project, where I learned the process of DNA barcoding and carried out an independent research project on biodiversity. This program sparked my first real interest in scientific research and led to my first peer-reviewed publication. Later as an undergraduate, I was selected for NIH Maximizing Access to Research Careers (MARC) program, a two-year honors fellowship to support students from diverse backgrounds into biomedical career pathways. MARC provided me with the mentorship, financial support, and research experiences that enabled me to transition directly from undergraduate to graduate school. These programs demonstrated to me how life-changing access to science education can be, and they instilled in me a deep commitment to pay this forward. As a PhD candidate at WashU, I have carried this passion into graduate school. I have taught and mentored students from high school to graduate levels through multiple programs. These roles taught me how to adapt complex scientific concepts into accessible, hands-on lessons that build student confidence and curiosity. They also reinforced my belief that mentorship and representation are critical for students to envision themselves as future

scientists.

At the same time, my research has given me an unique experience in virology, immunology, and advanced imaging. However the COVID-19 pandemic underscored for me that science communication is just as critical as scientific discovery itself. Effective science communication can positively shape public health outcomes and community responses. Coming from a background where I have seen how access to clear, accurate information can determine health and opportunity, I recognize the urgent need to equip students with not only scientific knowledge, but also the tools to evaluate and communicate it responsibly. For me, science communication is not only a professional responsibility, but a public health imperative, and this perspective prepares me to contribute meaningfully to Science Corps' mission.

I am drawn to Science Corps' vision of increasing equity in STEM education because it reflects my own lived experience. I know firsthand how access to mentorship and opportunities can change the trajectory of a student's life, and I am eager to help create those opportunities for others. By serving as a Science Corps Fellow, I hope not only to share my scientific knowledge and teaching experience, but also to learn from the communities I engage with, broaden my global perspective, and contribute to sustainable improvements in STEM education

In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.

In addition to my academic research in biomedical engineering, I have cultivated extensive experience in teaching, mentorship, and community engagement that will help me serve as a Science Corps Fellow. A unifying theme throughout my career has been using science as a tool for empowerment and equity.

At WashU, I have been deeply involved in mentorship and outreach at multiple levels. As a volunteer in the Young Scientist Program (YSP), I worked with high school students in under-resourced schools in St. Louis. YSP's mission is to attract students to scientific careers, particularly those historically excluded from STEM, by providing hands-on classroom demonstrations, lesson kits, scientific experiences, and mentorship. Through this role, I gained experience breaking down complex scientific concepts into interactive lessons that students could grasp, and I witnessed firsthand how exposure to engaging science activities can inspire curiosity and confidence.

I also served as a mentor in the McKelvey First gEN program, which supports first-generation undergraduate engineering students during their transition from high school to college. In this role, I mentored individual students one-on-one, providing guidance on both academic and personal challenges. This experience helped me build skills in peer mentorship and tailoring support to the unique needs of each student.

Within my graduate program, I have also taken on leadership roles focused on capacity building. As DEI and Peer Mentorship Chair on the Biomedical Engineering Doctoral Council, I designed and launched the first peer mentorship program for incoming PhD students. I secured funding, coordinated mentor-mentee pairings, and collaborated with faculty and administrators to integrate this program into the department. This initiative has helped me foster a stronger, more supportive environment for graduate students.

Most recently, I have consistently sought out opportunities to mentor, teach, and build community. As a McKelvey Engineering Graduate Ambassador, I serve as an official student representative for the School of Engineering,

where I support prospective, incoming, and current students. In this role, I share my experiences through virtual and in-person events. This work has strengthened my leadership skills, ability to communicate across diverse audiences, and commitment to making academic pathways more accessible.

These experiences have prepared me to thrive in a Science Corps Fellowship, where flexibility, creativity, and cultural humility are essential. I have learned how to adapt teaching methods for diverse audiences, create engaging science content from limited resources, and mentor students with varied backgrounds and goals. Just as importantly, I bring empathy, patience, and a strong commitment to equity in STEM. I believe these qualities, combined with scientific expertise, will enable me to contribute effectively as a Fellow.

Where do you imagine yourself professionally in the future? Check all that apply.

Doing science in industry

In science outreach

Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.

In the near term (within the next two years), I envision myself pursuing a career in industry or at a STEM-focused startup. After completing my PhD in Biomedical Engineering, I am eager to apply my training in virology, immunology, and bioengineering to the development of therapeutics, vaccines, or diagnostic tools. I am especially interested in contributing to translational research environments where innovative science can be rapidly converted into real-world solutions that improve health outcomes. Working in industry or an entrepreneurial setting will allow me to gain experience with product development, regulatory strategy, and collaborative problem-solving across disciplines.

In the long term, I aspire to create my own biotechnology startup or non-profit organization focused on addressing health disparities and expanding scientific capacity in underserved regions. My vision is deeply influenced by my family's roots in Guyana, where many communities face limited access to both healthcare and advanced education. I would like to leverage my scientific expertise, entrepreneurial training, and mentorship experience to build programs or companies that not only develop biotechnologies tailored to the needs of these populations, but also train and empower local scientists and educators.

In many ways, this aspiration mirrors the mission of Science Corps: creating opportunities for students in resource-limited settings to engage in meaningful science. I see this fellowship as a critical step towards my long-term vision. By teaching, mentoring, and collaborating with educators through Science Corps, I will develop cross-cultural communication skills, resourcefulness in teaching with limited infrastructure, and a deeper understanding of how to foster sustainable scientific capacity.

Ultimately, my goal is to bridge cutting-edge science with equitable access and education. In the short term, this means learning the operational and translational side of science through industry and entrepreneurship. In the long term, it means building something of my own. An organization that reflects both my scientific expertise and my personal commitment to creating opportunities for the next generation of scientists, especially in regions like Guyana.

Please upload your CV (.pdf) [Kuppan_J_CV_2025.pdf](#)**How did you find out about Science Corps?**

An email list in your institution

[Science Corps](#)